MACS+ - Membrane Action Of Composite Structures In Case Of Fire

R E P O R T

ArcelorMittal

1. Project properties

Project name:	Atlantic Park Phase 2 Unit 6	Prepared by:	KevinKurniawan
Client name:		Company name:	Fire Dynamics Group Limited
Client company:	Winvic Construction Limited	Date:	20 August 2025, 13:01
Job number:	0563	Checked by:	
Comments:	First and Second Floors Slab		

2. General arrangement

- Spans
  - Span 1: 7.1 m
  - Span 2: 9 m
- Unprotected beams
  - Number of internal unprotected beams: 2

3. Deck details

- Deck properties

Deck:	Multideck 60	Type:	Trapezoidal
Depth:	60 mm	Top flange:	142 mm
Pitch:	323 mm	Bottom flange:	119 mm
Stiffener height:	9 mm		

4. Slab details

- Concrete

Concrete type:	Normal	Slab depth:	150 mm
		Cylinder compressive strength of concrete (f_{ck}):	30 N/mm²

- Mesh

Mesh type:	A193		
Longitudinal mesh area:	193 mm²/m	Bar size:	7 mm
Transverse mesh area:	193 mm²/m	Bar size:	7 mm
Average mesh axis distance:	40 mm	Mesh yield stress:	500 N/mm²

5. Beams details

- Unprotected beams

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
		Degree of shear connection:	80 %
Section size:	UB 356x127x33		
Details:	h = 349 mm, b = 125.4 mm, t_w = 6 mm, t_f = 8.5 mm		

- Side A perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 356x127x33		
Details:	h = 349 mm, b = 125.4 mm, t_w = 6 mm, t_f = 8.5 mm		
Beam location:	Internal beam	Construction type:	Composite
		Degree of shear connection:	80 %

• Side B perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, t_w = 7.6 mm, t_f = 10.9 mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

• Side C perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 356x127x33		
Details:	h = 349 mm, b = 125.4 mm, t_w = 6 mm, t_f = 8.5 mm		
Beam location:	Internal beam	Construction type:	Composite
		Degree of shear connection:	80 %

• Side D perimeter beam

Beam type:	Solid beam		
Section family:	UK sections	Steel grade:	S355
Section size:	UB 457x152x52		
Details:	h = 449.8 mm, b = 152.4 mm, t_w = 7.6 mm, t_f = 10.9 mm		
Beam location:	Edge beam	Construction type:	Composite
		Degree of shear connection:	80 %

6. Loading details

• Normal (Cold)

Leading variable action:	2.5 kN/m²
Accompanying variable action:	0 kN/m²
Dead load including beam, excluding slab:	2 kN/m²
Calculated slab weight including mesh:	2.83 kN/m²

• Fire (Hot)

Combination factor for permanent action:	1.0
Combination factor for leading variable action:	1
Combination factor for other variable action:	1

7. Fire & Analysis

• Parametric temperature-time curve

Compartment length:	53.2 m	Window height:	3.8 m
Compartment width:	7.2 m	Window length:	53.2 m
Compartment height:	3.8 m	Glazing breakage:	80.011 %
Fire load:	321.83 MJ/m²	Wall lining (B) factor:	1700 J/m²s½K
Growth rate:	Medium	Combustion factor:	0.8

8. Summary output

• Default mesh direction

Maximum unity factor:	0.27	**Floor slab adequate**
Factored load in fire:	7.33 kN/m²	
Fire curve:	Parametric temperature-time curve	

9. Default mesh direction

• Tabular results

Longitudinal mesh area:	193 mm²/m	Bar size:	7 mm
Transverse mesh area:	193 mm²/m	Bar size:	7 mm

Factored load in fire: 7.33 kN/m²

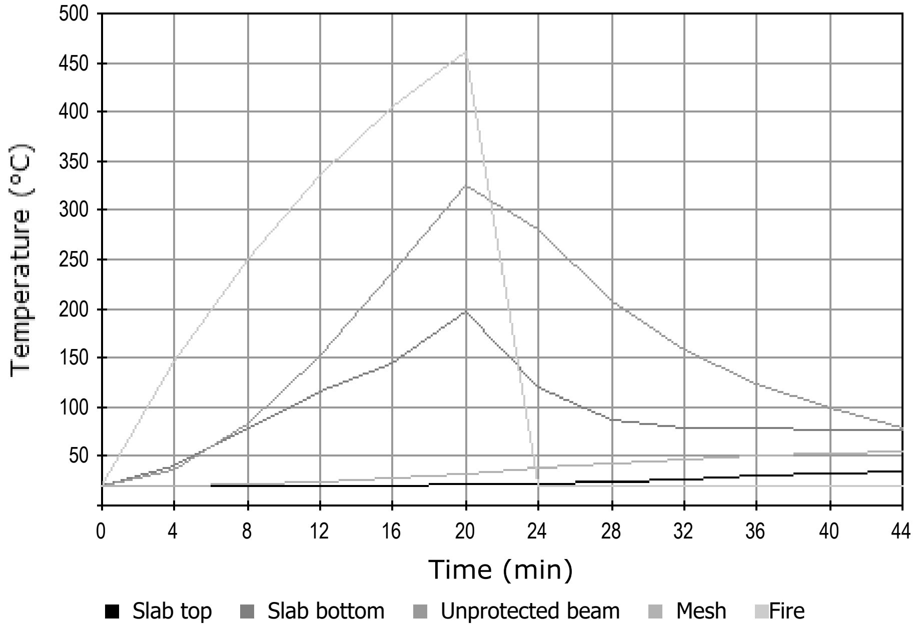
Time	Beam	Mesh	Slab top	Slab bottom	Beam capacity	Maximum allowable deflection	Slab yield	Enhance-ment	Slab capacity	Total capacity	Unity factor
mins	°C	°C	°C	°C	kN/m²	mm	kN/m²		kN/m²	kN/m²	
0	20	20	20	20	23.88	190	1.41	2.07	2.92	26.80	0.27
4	37	20	20	41	23.88	196	1.41	2.10	2.96	26.84	0.27
8	83	21	20	80	23.88	206	1.41	2.16	3.05	26.93	0.27
12	152	24	20	117	23.88	216	1.41	2.22	3.13	27.01	0.27
16	236	28	21	143	23.88	223	1.41	2.26	3.19	27.06	0.27
20	325	33	22	198	23.88	237	1.41	2.34	3.30	27.18	0.27
24	280	39	23	120	23.88	216	1.41	2.22	3.13	27.01	0.27
28	208	43	25	87	23.88	207	1.41	2.16	3.05	26.93	0.27
32	159	47	27	80	23.88	204	1.41	2.15	3.03	26.91	0.27
36	124	50	30	78	23.88	203	1.41	2.15	3.02	26.90	0.27
40	99	53	32	78	23.88	202	1.41	2.14	3.02	26.90	0.27
44	80	56	34	78	23.88	202	1.41	2.14	3.01	26.89	0.27

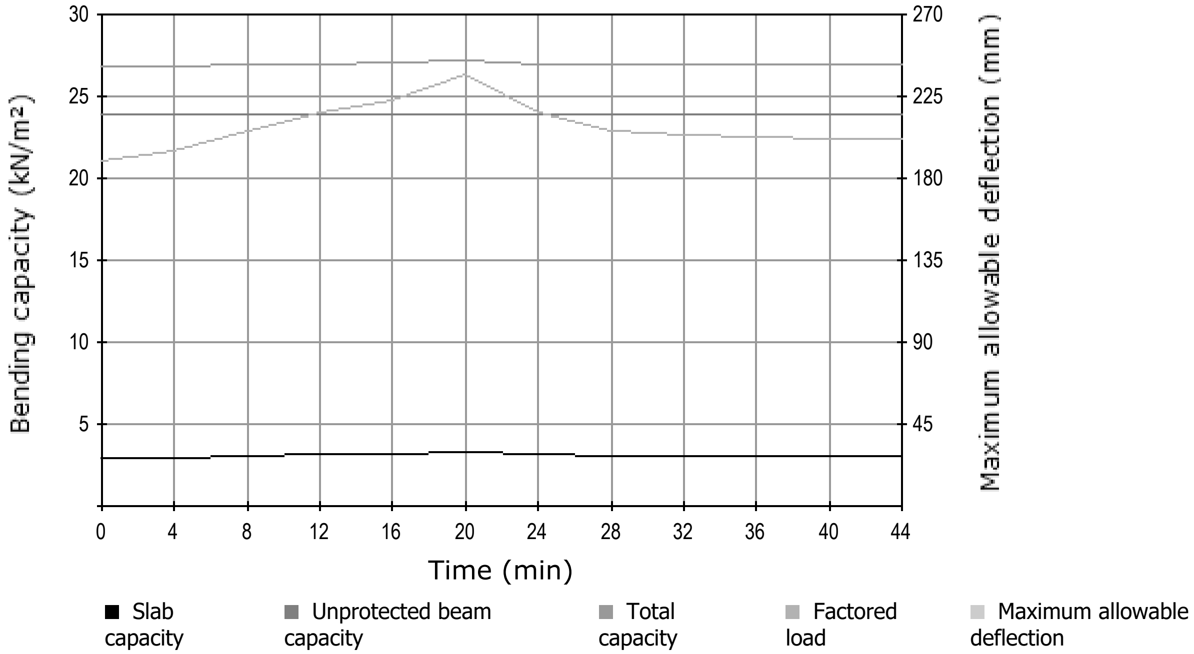
Maximum unity factor: 0.27 **Floor slab adequate**

- Perimeter beam check

Side A	Beam type:	Solid beam	Composite	Internal beam
	Section size:	UB 356x127x33		
	Required moment resistance in fire situation:	0 kNm		
	Line load in fire situation:	0 kN/m		
	Shear connection:	80 %		
	Degree of utilization:	0.71		
	Critical temperature:	544 °C		
Side B	Beam type:	Solid beam	Composite	Edge beam
	Section size:	UB 457x152x52		
	Required moment resistance in fire situation:	0 kNm		
	Line load in fire situation:	0 kN/m		
	Shear connection:	80 %		
	Degree of utilization:	0.71		
	Critical temperature:	541 °C		
Side C	Beam type:	Solid beam	Composite	Internal beam
	Section size:	UB 356x127x33		
	Required moment resistance in fire situation:	0 kNm		
	Line load in fire situation:	0 kN/m		
	Shear connection:	80 %		
	Degree of utilization:	0.71		
	Critical temperature:	544 °C		
Side D	Beam type:	Solid beam	Composite	Edge beam
	Section size:	UB 457x152x52		
	Required moment resistance in fire situation:	0 kNm		
	Line load in fire situation:	0 kN/m		
	Shear connection:	80 %		
	Degree of utilization:	0.71		
	Critical temperature:	541 °C		

10. Graphical output





MACS+ - Membrane Action of Composite Structures in Case of Fire. Version 3.0.4